

Company: MSCI

Job Profile: Technology and Data Operations Intern

Offer Type: Internship with Performance-Based PPO

Internship Stipend: ₹60,000 per month

CTC : ₹14.68 LPA (Base: ₹13.2 LPA)

Location: Mumbai / Pune

Process Date: Online Assessment – 11th July 2025, Interviews – 15th July 2025

Resume used -  Shubham_Karampure_Resume.pdf

Placement Process:

OA: (75 minutes) (~240 Candidates appeared)

This was an on-campus online assessment conducted on HackerRank, consisting of three questions: two DSA-based and one SQL. The questions were randomly selected from a pool, with most being LeetCode easy to medium level.

- DSA Q1: [Slowest Key – LeetCode](#)

Given a list of keypress logs as pairs (key, time), where key is an integer from 1 to 26 representing lowercase letters (1 → 'a', 2 → 'b', ..., 26 → 'z') and time is the cumulative time (in seconds) when the key was released, determine which key took the longest to press. The time taken for the first key is time[0], and for each subsequent key is time[i] - time[i-1]. Return the character that took the maximum time. It is guaranteed that this key is unique.

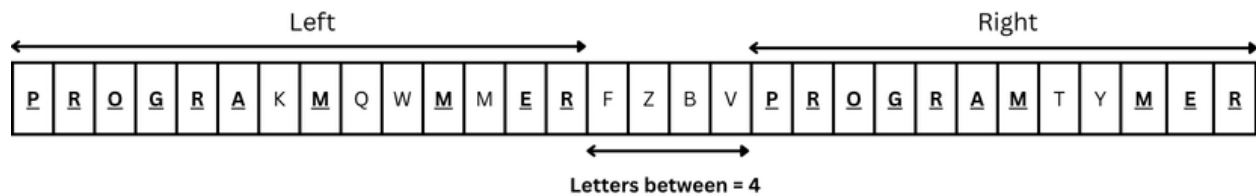
It's an easy single loop question, give it a try !

- DSA Q2: Distance Between "programmer" Substrings

Given a string containing exactly two non-overlapping occurrences of the word "**programmer**", possibly separated by arbitrary characters (e.g., "programxxxmerxxxxprogramxxmer"), determine the **number of characters strictly between** the first full "programmer" from the **left** and the last full "programmer" from the **right**.

Characters between may include any letters or symbols.

Example shown below -



I did using unordered_map and two pointers, give it a try !

- SQL Q3: Time and Amount Aggregation with Joins

The question involved 2–3 tables and tested a strong understanding of JOINS, GROUP BY, and aggregations. It required summing durations in seconds and converting them to HH:MM:SS format using functions like SEC_TO_TIME, as well as converting cumulative amounts into billions. I chose to skip this question as I wasn't aware of the SEC_TO_TIME function at the time. However, such functions can often be discovered if you forget using IntelliSense by exploring related keywords.

My strategy was to solve DSA quickly, assess the SQL difficulty, and submit if it seemed too complex — since submission time mattered and very few were able to solve SQL. This approach only works if you're well-prepared; don't skip doable questions.

Offline Technical Round 1: (30 minutes) (~40 Candidates appeared)

The round started with my introduction. I had prepared it well, covering languages I know, brief overviews of my three projects, extracurriculars, and leadership experience.

Tip: *A strong intro is important—it sets the tone and flow of the interview. Practice it in front of someone beforehand, to make it smooth and confident.*

She then asked about my DDPM (Denoising Diffusion Probabilistic Models) project. Since she wasn't very familiar with it, I explained it clearly using simple terms and a diagram, which she appreciated. Next, we discussed my first project which was an AI-assisted learning platform. I explained the idea, how I identified the problem, and the tech stack I used.

After that, she gave me two SQL questions.

Q.1 Find user IDs of users who have ordered products from more than one category.

Products table

product_id	product_name	category
1	Laptop	Electronics
2	Headphones	Electronics
3	T-shirt	Clothing
4	Book	Books

Orders table

order_id	user_id	product_id
1	101	1
2	101	3
3	102	2
4	103	4
5	101	2

I then explained that by grouping the results by user_id, I could use a HAVING clause with a COUNT(DISTINCT category) to filter for users who have ordered from more than one category. After I wrote the complete SQL query based on this approach, the interviewer was satisfied

Q.2 For each category, find the product that was sold the most (by total quantity ordered).

Products Table

product_id	product_name	category
1	Laptop	Electronics
2	Headphones	Electronics
3	T-shirt	Clothing
4	Book	Books

Orders Table

order_id	product_id	quantity
1	1	2
2	2	5
3	3	3
4	2	4
5	1	1
6	4	6

I shared my intuition to first group the orders by product to get the total quantity sold. Then, to use a window function to rank the products within each category based on their total sales. Finally, filter to get the top-ranked product per category. She was satisfied with the approach, and didn't expect to write the entire query.

After the SQL round, she asked two easy DSA questions:

Q1. [First Non-Repeating Character - Leetcode](#) I first explained the brute-force approach using two loops, then optimized it using a frequency vector to count frequencies and a second pass to find the first non-repeating character.

Q2. [Climbing Stairs - Leetcode](#) I related it to the Fibonacci series. Explained the recursive and DP approaches, but implemented the space-optimized (*Thanks to striver*) solution using two variables.

She also asked a few theory questions like encapsulation in OOP and indexing in databases, which I answered confidently with examples. At the end, I asked a prepared question related to the Barra product and RiskMetrics, she was working on it so she explained it well, and the interview concluded.

Offline Technical Round 2: (45 minutes) (~15 Candidates appeared)

The second round focused on evaluating my ability to think practically as a developer—beyond just coding skills.

We began with a short introduction, followed by a question about my future plans—whether I intended to pursue a master's degree or launch a startup. Since I had been part of the Entrepreneurship Cell at SPIT for two years. I responded honestly, stating that while I admire both academia and entrepreneurship, my goal is to grow in an industry setting.

Given my background in generative AI, he asked : *With AI advancing rapidly, what is the role of a developer today?* I highlighted that while AI has made remarkable progress, it still has significant limitations. It lacks genuine reasoning and primarily functions as a predictive model trained on past data. More importantly, AI cannot take accountability. Developers remain crucial for testing, debugging, refining logic, and owning the final product.

We then dove deep into my projects. I presented a generative AI-based application, clearly outlining the problem it solved, the tech stack used, and design decisions. A major discussion point was my use of MongoDB instead of a traditional SQL database. I explained that since LLMs can output structured JSON, it was easier and more efficient to store and manage data in MongoDB. The project didn't require complex joins or strict ACID compliance, and MongoDB's horizontal scalability and high performance made it a practical choice.

He also asked about the vector database I had used—Qdrant. I elaborated on why I chose it, highlighting its ability to perform advanced filtering during semantic search. I gave concrete examples from my project to justify the choice, and the conversation naturally progressed into the architecture, challenges faced, and how I overcame them.

Throughout the discussion, the focus remained on *why* I made certain technical decisions. We also covered theoretical questions—like the difference between a database and a data warehouse, and what an alternate key is. Rather than just naming types of keys, I walked him through the concept of alternate keys within a hierarchy of super, primary, candidate, and foreign keys.

Toward the end, he asked me about time complexity. Instead of giving definition, I explained it through real-world examples—demonstrating how analyzing time complexity helps identify performance bottlenecks across different hardware and programming languages.

One of the final questions was : *Would you choose expensive infrastructure to reduce server load time, or stick to a budget and accept slower performance?* I treated it as a real-life scenario. I said that in a perfect world with unlimited resources, we'd all want the best infrastructure—but in reality, we have to balance performance and cost. For instance, if I'm building an app like YouTube, users won't wait 20 seconds for a video to load. In such cases, performance becomes a priority. So I'd first identify the bottlenecks and then decide where to optimize for speed and where to save on budget—always keeping both business constraints and user experience in mind. He seemed satisfied with this practical outlook.

As the interview wrapped up, he asked if I had any questions. I asked about their adoption of generative AI. We had an engaging discussion on how they're exploring AI integration—whether through external APIs or on-premise LLMs. I also raised concerns around data security when relying on third-party APIs, especially for data-sensitive businesses. He acknowledged that the domain is still evolving, and while they're already using AI tools for tasks like code assistance, larger implementation plans are in the works.

Tip: *In college or hackathon projects, we often neglect the feasibility and scalability aspects. It's still important to have a basic theoretical understanding of these, even if you don't have practical experience*

Online HR Round : (15 minutes) (~8 Candidates appeared)

This was a relaxed and conversational round, primarily focused on understanding my personality and cultural fit. The interviewer began by asking about my family background and hobbies. I had prepared a brief HR-friendly introduction that covered my academic journey, family background, and my passion for playing musical instruments. He then asked what I knew about MSCI, to which I gave a detailed response—covering the company's role in global finance, its significance in investment decision-making, and how MSCI indices provide a global snapshot to help investors make informed choices. I also touched on the technical side of index construction. He expressed curiosity about my specialization in Data Science, and asked whether I'd be open to working in a different domain or department. I explained how my foundation in data science and my keen interest in generative AI have equipped me with strong analytical and adaptable skills. I assured him that I'm quick to learn and flexible enough to contribute effectively, regardless of

the specific domain. Towards the end, I asked a couple of questions—particularly about the scope of learning before the internship begins, since we had a semester gap, and inquired about growth opportunities within the company.

Overall, it was a smooth and engaging conversation that allowed me to present both my technical background and willingness to learn and grow.

Tip: Don't overlook the HR round. Be well-prepared with common HR questions and think through different scenarios in advance. You won't have much time to recollect answers during the conversation, as this round often tests your spontaneity and presence of mind

At the end, 5 students were selected.

Useful Tips:

In General -

1. Placement is 30% luck, 70% smart work. Focus on what you can control—be well-prepared when the opportunity comes.
2. Give contests and mock interviews even if you feel underprepared. They help you assess your level and improve.
3. Don't fear early failures; better to fail in practice than in interviews.
4. Focus on one company at a time, give your best, and if it doesn't work out—learn, move on, and stay resilient.

These may sound simple, but I struggled to follow them myself at first—don't repeat that mistake.

For MSCI -

1. Research the company thoroughly—memorize key facts & values but also understand their meaning and relevance to your profile.
2. Soft skills are crucial. Even if my answers weren't perfect, I spoke with confidence, used technical terms, and explained with clear, relatable examples.
3. For almost a month, I reviewed past interview experiences and practiced answering questions out loud, refining them for clarity and impact

Preparation Resources

Category	Resource/Platform	Key Strategy/Notes
OA - Aptitude	IndiaBix	Questions are straightforward and frequently repeated. Focus on revising formulas and practicing key topics (percentages, ratios, time-distance).
DSA	Striver's A2Z DSA Sheet	Prioritize understanding patterns and logic over memorization. Covers almost all major company patterns and builds strong problem-solving skills.
Core Subjects	Pick any YouTube Lectures & GFG Notes	Watch one shot, complete-topic lectures for efficient learning and take handwritten notes. Use GFG's Last Minute Notes for quick revision.
↳ DBMS	GFG DBMS Notes	Supplement lecture notes with these for fast review.
↳ CN	GFG CN Notes	Supplement lecture notes with these for fast review.
↳ OS	GFG OS Notes	Supplement lecture notes with these for fast review.
AI Tools		
↳ Recall	Gemini	Use its voice assistant feature to simulate quick mock interviews on technical topics for active recall practice.
↳ Research	Perplexity	A fast, reliable tool for researching company backgrounds, recent news, tech stacks, and interview experiences.

ALL THE BEST FOR YOUR PLACEMENT !!

Feel free to reach out to me on LinkedIn if you have any questions or need any help—I'll be happy to support you

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